Assessment of Factors Influencing the Volume of Personal Income Tax Revenues

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***Abstract*— Individual income tax is the leader among tax revenues. The completeness of its revenues directly depends on the situation in the country. In conditions of instability, in order to preserve their welfare, taxpayers are forced to evade taxation or minimize the amount of payment. The aim of the study is to assess the impact of macroeconomic indicators on personal income tax revenues. Identification of the factors of positive impact on the receipt of personal income tax will improve planning of its volume in the future. The research is based on the results of activities and characteristics of the labor resources of Ukraine, which directly affect the base of taxation of individual income. By means of IBM SPSS STATISTICS software product statistical analysis and time series analysis has been carried out. The applied analytical program Statgrafhics was used for conducting correlation and regression analyses. Statistical data of the State Statistics Service and operational data of the Tax Service of Ukraine became the information base for the research. The work carried out a comparative analysis of real and nominal incomes of the population of Ukraine and the income tax for eighteen years. It was determined that in comparison with 2001 the real value estimation decreased by 7.13 times. Average annual growth of personal income and tax volumes differ significantly, which indicates partial taxation of income. Statistical analysis of indicators indicates a normal distribution, which allowed to make a short-term forecast of the following. The impact of factors on tax revenues during a period of stability and political crisis was assessed. The correlation analysis showed that the level of unemployment, consumer price index and the number of informally employed people do not affect the income tax receipts for individuals. It should be noted that the number of informally employed population in the regions of Ukraine is 22.4%. Employers do not levy any tax on these citizens and the budget loses funds, so there is no connection.**

Keywords*— real and nominal tax on personal income, correlation and regression analyses, personal income, employed population, labor demand, average monthly wage, unemployment.

I. Introduction (*Heading 1*)

In order for the State to perform its functions, it needs financial resources, the main source of which is tax and non-tax revenues. In 2019 the volume of tax revenues in the consolidated budget of Ukraine amounted to 1050,9 billion UAH, which is almost 90% of the total. The main source of tax revenues is three national taxes, which in 2019 accounted for almost 45% of the total amount. Individual income tax (25.7%), corporate income tax (10.04%) and value added tax (9.00%) take precedence [7]. The voluntary payment of taxes is negatively affected by the financial, economic, political crises and the global pandemic. In conditions of instability, there are more cases of tax evasion, and legal and illegal payments are minimized. Therefore, the assessment of the factors of positive impact on the receipt of tax on personal income will improve the planning of its volume for the future.

The issue of taxation of personal income has received considerable attention. Let's consider recent research. Dutova N. V. and Lesik E. S. determine the advantages and disadvantages of taxation of personal income in Ukraine and abroad. They assess the possibilities of using and implementing the experience of EU tax reforms in Ukraine [4]. Lobodin Z. M. analyses the role of the personal income tax in the formation of budgetary funds and its impact on the socio-economic development of the state. Identifies problems of transfer of the specified tax, its distribution between budgets and indemnification of expenses of the physical persons, attached to the tax discount on the tax on incomes of physical persons [6]. Zadorozhnya L. A. developed a methodological approach to the calculation of the tax wedge of taxpayers with different social status by adapting the OECD (The Organization for Economic Cooperation and Development) methodology to the conditions of Ukraine.

It proves that the regulatory effectiveness of personal income tax is not enough, because the differentiation of tax wedges by the status of the taxpayer is insignificant, and the coefficient of tax participation shows that in conditions of receiving social payments there may be no incentives to work [5]. Gaidai V. I. proposes to introduce a mechanism of progressive scale of income taxation from 18% to 40%, which will increase the tax burden on taxpayers with high income level. To establish an untaxable minimum income from which the tax on personal income will not cope and leave the tax proceeds in the local budgets. Quantitative assessment of personal income tax with the help of economic and mathematical methods and models was left out of sight of researchers, which caused the urgency of this research.

The purpose of the study is to determine the factors influencing the volume of personal income tax receipts, assessment of labor resources as a source of taxation. In order to achieve the goal of the research, statistical, correlation, regression and time series analysis are used. The statistical data of the State Statistics Service and operational data of the Tax Service of Ukraine became the information base for the research. The applied analytical program Statgrafhics and SPSS were used for conducting correlation and regression analyses.

II. MAIN PARTY

When analyzing value indicators over a long period of time, the impact of inflationary processes should be taken into account. It is offered to consider dynamics of receipt of the tax on incomes of physical persons and the general incomes of the population across Ukraine in prices 2001. Cumulative consumer price index calculated by the formula [1] was used to calculate the real indicators:

$$R_i = \prod_{j \le i} \left(1 + r_j / 100\right), \quad (1)$$

r_j - annual consumer price index in %, $\prod$ - defines the product of the expression. The analysis of the results showed that over seventeen years the real value estimation of indicators decreased by 6.85 times (tab. 1).

TABLE I. **DYNAMICS OF INCOMES OF THE POPULATION OF UKRAINE AND INCOMES OF THE TAX ON INCOMES OF INDIVIDUALS FOR 2002-2019**

Year	Consumer price index,%	Cumulative index*	Nominal incomes of the population of Ukraine, million UAH	Nominal personal income tax (NPI), million UAH	Real incomes of the population of Ukraine, million UAH	Real personal income tax (RPI), million UAH
2002	99,40	0,99	185073,00	10847,00	186190,14	10912,47
2003	108,20	1,08	215672,00	13548,00	200530,35	12596,84
2004	112,30	1,21	274241,00	13241,00	227059,14	10962,95
2005	110,30	1,33	381404,00	17346,00	286296,69	13020,58
2006	111,60	1,49	472061,00	22820,00	317515,56	15349,09
2007	116,60	1,73	623289,00	34812,00	359548,77	20081,55
2008	122,30	2,12	845641,00	45928,00	398866,82	21663,04
2009	112,30	2,38	894286,00	44515,00	375611,22	18696,85
2010	109,10	2,60	1101175,00	51057,00	423929,58	19655,89
2011	104,60	2,72	1266753,00	60227,00	466227,23	22166,49
2012	99,80	2,71	1457864,00	68092,00	537640,73	25111,42
2013	100,50	2,73	1548733,00	72151,00	568310,45	26475,94
2014	124,91	3,40	1516768,00	75203,00	445585,49	22092,61
2015	143,32	4,88	1772016,00	99983,00	363222,48	20494,21
2016	112,42	5,48	2051331,00	138782,00	374022,08	25304,32
2017	113,72	6,24	2652082,00	185686,00	425217,96	29771,71
2018	109,82	6,85	3248730,00	229901,00	474304,05	33564,80
2019	104,10	7,13	3699346,00	264440,70	518820,87	37086,92

Source: authors own calculations based on data [1, 3, 8].

The analysis of time series indicates that during eighteen years nominal incomes of the population of Ukraine increased by 183.5 billion UAH on average. At the same time, the real average annual growth rate is 11 times less - 16.2 billion UAH (tab. 2).

TABLE II. **REAL AND NOMINAL DATA FOR 2002-2019, MILLION UAH**

Indicator	Equation	Determination coefficient
Nominal incomes of the population of Ukraine	Y = 183535x - 398778	$R^2 = 0{,}8998$
Real incomes of the population of Ukraine	Y = 16194x + 232211	$R^2 = 0{,}6036$
Nominal personal income (NPI)	Y = 12716x - 40329	$R^2 = 0{,}7972$
Real personal income (RPI)	Y = 1285,2x + 9180,3	$R^2 = 0{,}8547$

Source: authors own calculations based on data [1, 3, 8].

Absolute average annual growth of nominal receipts of personal income tax is 12.3 billion UAH, and real value of funds is 9.9 times less. Annually, for 2002-2019, the budget additionally received 1.3 billion UAH. The discrepancy between the increase between the income of population and income tax (1,6) attracts attention. Attention is drawn to the mismatch between income and income tax: (12,7 * 100% / 183,5 = 6,9%). The main personal income tax rate is much higher. It is expedient to study in further studies the volumes and types of benefits provided, which exempt income from taxation. Also, to differentiate income by type, they have different tax rate.

Real incomes of the population of Ukraine for 2002-2019. On average, they amounted to 385.05 billion UAH, the minimum income was in 2002 (186.19 billion UAH), the maximum in 2013 (568.31 billion UAH). The scope of variation is 382.12 billion UAH. There is a normal distribution in the distribution (result <2):

kurtosis / standard kurtosis error = -0.304 / 0.536 = 0.57, (2)

asymmetry / standard asymmetry error = -0.549 / 1.038 = 0.53. (3)

A short-term forecast can be made. Thus, in 2020 real incomes of the population will vary from 330.72 to 441.38 billion UAH.

Similarly, let us calculate the forecast for personal income tax (tab. 3).

TABLE III. **FORECAST OF REAL AND NOMINAL INDICATORS FOR 2020, BILLION UAH**

Indicator	95% confidence interval average	
	lower bound	*upper bound*
Nominal incomes of the population of Ukraine	831,15	1858,46
Real incomes of the population of Ukraine	330,72	441,38
Nominal personal income (NPI)	42,67	118,29
Real personal income (RPI)	17,70	25,08

Source: authors own calculations based on data [1, 3, 8].

Statistical estimates indicate a normal distribution for other indicators as well.

Let us estimate the impact of macroeconomic indicators on the amount of personal income tax collected (Y). For this purpose, we apply a correlation analysis. The initial data for the analysis are the following indicators for 24 regions of Ukraine in 2018: X_1 - employed population, thousands of people, X_2 - demand for labor, thousands of people, X_3 - resident population, thousands of people. Person, X_4 - average monthly salary, UAH, X_5 - level of registered unemployment, in % of working age population, X_6 - disposable income per person, UAH, X_7 - consumer price index, %, X_8 - income of population, millions of UAH, X_9 - number of informally employed population.

Pearson's correlation matrix has been calculated, which indicates that the unemployment rate (-0.306), consumer price index (0.232) and the number of informally employed population (0.104) do not influence the income tax receipts of individuals (tab. 4).

TABLE IV. **PAIRS OF PEARSON'S CORRELATION COEFFICIENT MATRIX**

Factors	1	2	3	4	5	6	7	8	9	Y
1	1									
2	0,731	1								
3	0,723	0,411	1							
4	0,542	0,394	0,723	1						
5	-0,440	-0,305	-0,477	-0,277	1					
6	0,608	0,581	0,028	0,284	0,123	1				
7	0,214	0,003	0,489	0,435	-0,493	-0,284	1			
8	0,968	0,720	0,811	0,660	-0,383	0,571	0,213	1		
9	0,322	0,033	0,055	-0,065	-0,265	0,238	-0,299	0,229	1	
Y	0,933	0,776	0,777	0,708	-0,306	0,601	0,232	0,974	0,104	1

Source: authors own calculations based on data [1, 3, 8].

In 2018, the average registered unemployment rate for the working age population is only 2%, so the impact is low. The average consumer price index in 2018 is 110% and this indicator has an average connection only with three indicators, registered unemployment rate (-0.493), number of permanent population (0.489) and average monthly salary (0.435).

In terms of the number of informally employed population, their average number in the region is considerable 141 thousand people, which is another 22.4% of the additionally employed population. According to employers, they do not levy any tax on these citizens and the budget loses funds, so there is no connection. All other factors have a strong direct link (more than 0.7), only the average income per person (0.601).

Matrix analysis indicates the presence of multicollinearity between indicators. For further regression analysis, it is necessary to apply the Ridge Regression, which allows to differentiate between the influence of factors on the performance attribute.

The estimates of Variance Inflation Factors (hereinafter VIF) confirm the presence of multicollinearity in the model on four indicators calculated value greater than 10 (tab. 5).

According to Ridge Parameter 0.02, the value of VIF estimates are less than 10 for all factors (tab. 6).

TABLE V. **RIDGE REGRESSION RESULTS**

Parameter	Constant	X_1	X_2	X_3	X_4	X_6	X_8
Variance Inflation Factor	-6009.4	22,63	2,82	39,11	4,04	14,95	81,39

Source: authors own calculations based on data [1, 3, 8].

So we apply it to determine the parameters of the regression equation:

Y_{2018}=-6009,4 +1,08 X_1+434,57 X_2+1,29 X_3+0,53 X_4+0,07 X_6+0,02 X_8 (4)

The increase in the number of employed population per 1000 people will lead to the growth of tax revenues by 1.08 million UAH, while other factors are constant. Growth in demand for labor force per 1000 people will increase receipts by 434.57 million rubles, accordingly. An increase in the average size of the permanent population by 1 thousand people will increase revenues by 1.29 million UAH, respectively. The influence of incomes on tax receipts is much less: thus the growth of average salary by 1 UAH will lead to the increase of 0,53 million rubles, disposable income by 0,07 million UAH, and the growth of total incomes by 1 million UAH will increase incomes by 0,02 million UAH accordingly.

TABLE VI. **RIDGE REGRESSION RESULTS**

Ridge Parameter	X_1	X_2	X_3	X_4	X_6	X_8	R^2
Variance Inflation Factors 2018 year							
0	22,63	2,82	39,11	4,05	14,95	81,39	97,43
0,005	14,84	2,44	17,47	3,21	7,46	31,03	97,25
0,01	11,11	2,28	10,65	2,78	5,00	16,53	97,09
0,015	8,80	2,18	7,47	2,49	3,81	10,38	96,94
0,02	7,21	2,10	5,66	2,28	3,11	7,20	96,79
Standardized Regression Coefficients							
0,02	0,08	1,20	0,28	0,10	0,21	0,33	2018 year
0,02	0,11	0,01	0,17	0,19	0,01	0,52	2013 year

Source: authors own calculations based on data [1, 3, 8].

A qualitative regression model was built, which is indicated by the Fisher criterion 107 it is significant (p = 0). By 96.79% the resultant feature depends on the selected factors and by 3.21% of other values not used for the model construction, taking into account the error. The model has no autocorrelation calculated Darbin-Watson criterion 2.05.

Standardized Regression Coefficients allow the measurement of impact through a relative value. Thus, in 2018 the first place is occupied by the demand for labor force (1.2), the second position - by the income of the population (0.33), the third - by the average number of permanent population (0.28). The outsiders are the number of employed population (0.08) and the average monthly salary (0.10) (Table 3). The similar model was built based on the data of 2013 and has a completely different distribution by the relative impact on income tax receipts. Thus, the leader is personal income (0.52), the second place is occupied by the average monthly salary (0.19), the third place is occupied by the average number of permanent population (0.17).

The regression model is based on the statistics of 2013 and is adequate: Fisher 116 and Darbin-Watson 2.4 criteria, determination coefficient R2 = 96.35.

III. Conclusion

The obtained model according to the data of 2018 indicates that the development of industry in the domestic market increases the supply of employers and restrains the outflow of labor abroad. Salaries have a low impact, although they are the main source of taxation. Personal income has shown a much greater impact, indicating that the population receives different types of income, which are also taxed. In the period of stability (2013), tax revenues are affected by quite different indicators, namely, income, wages and number of permanent population. Revenues and wages directly reflect the base of the study, which is logical for the personal income tax. The military actions in eastern Ukraine have led to inflationary processes, loss of real incomes of the population, reduction of demand for labor force within Ukraine for decent wages. These events have led to labour migration outside Ukraine in search of decent wages.

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